Brief for Mr Abhishek Pitale — From Labour Care Documentation to Maternal Death

Review Seven-day assignment · Nov 2, 2026; Prepared By: Dr Pradeep Atter

Welcome and purpose

Over seven days, we would like you to explore — and demonstrate in a working prototype — whether information recorded while a woman is in labour can be safely reused when her death later has to be reviewed, so that a doctor does not have to reconstruct it.

The assignment tests how you think as a technical leader:

• how quickly you understand two unfamiliar clinical workflows and the relationship between them.
• whether you can see what overlaps, what does not, and what only *looks* as if it overlaps. • how you design for exhausted users under pressure.
• how you protect the accuracy, traceability and integrity of records that may later be examined closely.

The problem: from labour ward to death review

Doctors and nurses document clinical information during labour. If a maternal death occurs, they must complete an eight-page Facility-Based Maternal Death Review (FBMDR) form within 24 hours.

Maternal death is defined by the WHO as the death of a woman while pregnant or within 42 days of termination of pregnancy, irrespective of the duration or site of pregnancy, from any cause related to or aggravated by the pregnancy or its management, but not from accidental or incidental causes.

The FBMDR form requires patient demographics, obstetric history, referral details, labour and delivery records, interventions, complications, and circumstances surrounding the death. Much of this information has already been recorded in clinical documentation, including the Labour Care Guide (LCG).

Problem: Clinical information documented during labour is not automatically transferred to the FBMDR form. Healthcare workers must manually retrieve, verify, and re-enter the same information, resulting in duplicated work, increased documentation burden, and potential transcription errors.

Workflow A: labour documentation and the WHO Labour Care Guide: Provided as PDF

Workflow B: facility-based maternal death review: Provided as PDF

Design principles: provenance, integrity and clinical honesty

Capture once. Validate appropriately. Reuse where permitted. Report accurately.

Ensure the following:

1. **Every imported value is traceable.**
2. **The original is never changed by the review.**
3. **Authorship is preserved.**
4. **Contemporaneous and retrospective are kept apart.** Information recorded at the time is different

from information added after the death. Entries made to the labour record after the time of death should be clearly identifiable.

5. **Clinician's explicit decisions are respected.**
6. **Reuse is offered, not imposed.** The clinician sees what is proposed for reuse, accepts or rejects it, and is accountable for what goes into the review.
7. **Nothing is silently changed after submission.** Corrections are visible, attributed amendments.
8. **Reviews are confidential and for learning.** ***Design language and permissions should support honest reporting, not blame.***

Who you are designing for. A resident at 4 a.m., post-call, on phone, who has just been part of a failed resuscitation and still has other patients. They will be interrupted. They need to see at a glance what the labour record already supplies, what is still missing, and what does not apply — and to trust that what is reported is what was recorded.

We designed the UI for interruption, not for a calm desktop session. The resident should not have to remember where they stopped or reconstruct what the labour record already contains.

The first screen shows status before detail: progress, what is still missing, open proposals, and how many questions are hidden because they do not apply. Each value carries a small origin mark reused, derived or entered and a provenance link that opens the source entry, author and time. Missing observations stay visible as blank cells or dots; they are not silently filled.

Inputs are constrained. Fixed choices use selects or radio pills, dates and times use native controls, and blocks use fixed columns. Questions that cannot apply are hidden, and N/A hides its children. This prevents errors before they happen instead of showing a warning later.

Work autosaves after a short pause and on blur, with a visible last saved stamp. A resident can close the phone, return, and continue. Checks run only when the resident asks for them, so the form does not shout while they are still working. Errors block; warnings need one confirmation. Proposals show their source, derived values show the calculation, and alerts appear next to the relevant form field. Nothing is silently overwritten. Submitted and approved versions are frozen, and each printed value keeps its source chain.

What we want you to build: the demo journey

A working application, on fictional data, that holds labour records, lets a doctor start a death review from a woman's record, proposes what can be reused with its source visible, guides the doctor through everything else, and produces a structured review output that an FNO can approve — with nothing typed twice and nothing claimed that was not recorded.

The journey we expect to see in your demo

1. **Labour is documented.** A fictional woman's labour record exists, with observations entered over time by more than one person (for example a nurse and a resident), including at least one value crossing an alert threshold and at least one gap.

2. **She dies.** The time of death is recorded.
3. **A review begins.** A resident opens a new death review linked to her record.
4. **Reuse is proposed.** The system shows which review items it can fill from the labour record, how, and from which entries — and which it cannot.
5. **The resident decides.** They accept, reject or correct each proposed value; corrections do not alter the labour record.
6. **The rest is completed.** Everything the labour record cannot supply is captured through a guided,

conditional experience, resumable after interruption.
7. **Judgements are the resident's.** Cause of death, contributing factors and the narrative are entered as the clinician's own conclusions, with relevant labour observations shown alongside as evidence.
8. **Review and approval.** The resident submits; an FNO returns it with a comment; the resident fixes it; the FNO approves.
9. **Structured output.** The approved review produces an output suitable for maternal death review — structured like Form 4, and showing for each value whether it was reused, derived, or entered, and from where.

10. **A case without a labour record.** Show the same workflow for a woman who died before labour or was referred in after delivery.
11. **History.** Show who did what, and when, across both records.

How you get there is your decision. We are not specifying screens, how the labour record is entered, how reuse is presented, or how the review is structured. We are interested in why you made the choices you did.

Requirements: essential, desirable, deferred

Complete the essential list well before reaching for anything else. Tell us which desirable items you chose and why, and what you deliberately left out.

Essential

1. **Overlap analysis.** A written mapping of every Form 4 item to the four categories in "The core question", with reasons and with the differences in terminology or precision you noticed.

If we consider lcg as the sole form, only one key matches readily, that is the parity. But considering a few other documents, like the related patient records, hospital entries, etc, around 17 msdr keys become available. More info in docs/overlap.md

2. **Labour records.** Synthetic labour records structured like the LCG, with timestamps and the author of each entry. **You do not need to build a full LCG charting interface.** Seeded data plus a simple way to add or view entries is enough; spend your time on reuse, not on drawing the chart.

A simple LCG is attached to most of the msdr records, apart from the records that do not have a labor record. Any value in the Labor record can be added through the UI, and the appropriate alerts for the values in the labor records are also programmed in a way that a person adding values to the labor record cannot go beyond the thresholds. Also the person who adds those values is documented and the history is preserved. More info in docs/lcg-storage.md

3. **Review linked to the source.** A death review can be created from a woman's labour record — or without one.
When a death review has to be created, you get an option whether to add a label record to it or not to add a label record to it.

4. **Proposed reuse with provenance.** Values the labour record can supply are proposed, each showing its source entry, time and original author. Derived values show how they were calculated.
Each value that is added has a provenance to it.The proposed value has all the information as to who got that value there in the first place. Same with the derived values.

5. **Clinician acceptance.** Each proposed value is accepted, rejected or corrected by the clinician. The labour record itself is never changed by the review.
Each value that is proposed or derived can be accepted, rejected or corrected by the clinician.

6. **The rest of Form 4.** Everything else can be captured through a guided, conditional experience that hides what does not apply and can be resumed later.
Yes.

7. **No unsupported conclusions.** Cause of death, diagnoses and contributing factors are only ever chosen by a clinician. Relevant observations may be shown as evidence.
Yes.

8. **Gaps and late entries visible.** Missing observations are shown as missing. Entries added to the labour record after the time of death are identifiable.
Every entry in the labor Record can be edited. based on when the woman actually died and based on the entry, the program tells us whether the entry added was late or not.

9. **Checks before submission.** Missing, impossible or inconsistent information is caught and explained in plain language.
There are several checks for each section of the MSDR form that keep on happening as the Clinician is entering the information and before submission, a Clinician gets an alert if there are any checks that seem other than the usual. And the Clinician is guided through those checks in the same form itself. The Clinician can then decide to change the values based on the checks that he gets from the program or he can decide to submit it anyway and wait for the FNO approval. More info in docs/checks.md and docs/conditions.md

10. **Review and approval.** At least author (resident) and approver (FNO) roles, with return-with comments and approval; approved reviews are locked and later changes are attributed amendments.
Once the person submits the form, the FNO gets the form to either approve it or comment the field out which has an issue or comment the entire form and then return it to the clinician.

11. **Structured output.** An output suitable for maternal death review, following Form 4's structure, with the origin of each value available.
Each value's history is available and is stored correctly and the output is available in a very structured format during the form4 review.

12. **History.** Who created, changed, accepted, submitted, returned and approved what, and when.
There is a separate history tab for form 4 and LCG for looking at form changes

13. **Synthetic cases.** At least four: a death after a charted labour; a caesarean after a period of charted labour; a death with no labour record (before labour or after an abortion); and a woman referred in after delivering elsewhere.
All four cases have been recovered by the program.

Desirable — choose deliberately

• Form 1 produced from the same review with no re-entry.

Since Form 1 pretty much coincides with Form 4, apart from a few details, whatever coincides gets filled automatically, and the user is asked to fill the other items. But filling the form one is not mandatory to submit forms4.

• A timeline view combining labour observations and later events, distinguishing contemporaneous from retrospective entries.
Yes, values added to the L C G are separated as to when they were added, whether they were added before or after death occurred.
• Alert-threshold crossings from the labour record surfaced as evidence next to the relevant review items.
All the values in LCGs have thresholds which have been programmed in and you cannot enter values which are outside the threshold.

• A second labour-record format (a partograph or a hospital variant) mapped without changing the review model.
Decided not to have a very specific second lea second lever record format but decided on a few other smaller types of information that we could get from our various other hospital sources but that's general information with respect to the patient.

• Printable Form 4 layout suitable for a wet signature.
Yes, this is ready and available when the form is filled and approved.

• Two hospitals in one deployment, each unable to see the other's records.
I decided not to do that because it doesn't really suit the problem statement and adds to the complication.
• Automated tests for mapping, derivation and validation rules; one-command local setup.
Tests are written for every section in the form, and the app has a one command setup. More in readme.md

Deferred — please do not build

• A full graphical LCG with plotting and alert logic.
• Connections to any government or hospital system (MPCDSR portal, HMIS, ABDM, hospital information systems).
• Issuing death certificates.
• A committee module with scheduling and notifications.
• Offline synchronisation (discuss it in your documents if you think it matters).
• AI that writes clinical narratives or suggests causes of death.
• Production deployment or compliance certification.

omitted all of this.

Technical freedom and open-source preference

Complete technical freedom. **We strongly prefer established, actively maintained open-source technology and avoid closed source dependencies.** Include a short **dependency register**: each significant component, its licence, why you chose it, and its maintenance status.

We have decided to go with python, django and postgresql for this. All three are very reliable and maintained. More info in docs/security.md and docs/dependencies.md

If you use a health-data standard (for example HL7 FHIR resources for observations), explain what it buys you and what it costs in seven days. You are not required to use one.

Decided not to go with any of the health standards, reason mainly being that, it doesn't really adhere to the crux of the assignment, which is making death review form reusable. If needed, we can very easily implement this in the future.

Security and data integrity

Keep in mind integrity as well as confidentiality. Keep in mind the DPDP rules.

These are the major rules the app follows wrt dpdp:

- Purpose limitation: the app only collects information and data thats needed (form 4)
- Strong accuracy: the app is append only. Corrections need a new row and does not change the existing db row.
- Login is required on every page
- There is auth scoping to resident and fno. Auth works on the principle of least privilege. Backups are encrypted. No ai, data sharing or external analytics.
- Data is encrypted in transit and in rest. In postgres via scram, and in transit via caddy, with optional tls.

Data model expectations

Choose whatever database and modelling approach you think fits, and explain why.

I used PostgreSQL 17 with the Django ORM. The model separates four things: recorded facts, clinician answers, reuse proposals, and frozen versions. Facts such as admissions, deliveries, deaths and LCG observations are normalised rows. They are append only and corrected by a new row with supersedes_id. Each printed Form 4 field has one ReviewItem, so one answer per field is enforced by a unique constraint. Repeating blocks use schema-validated JSONB because their columns differ. Submitted and approved outputs are frozen JSONB payloads with SHA-256 hashes and per-value provenance.

PostgreSQL fits because the app needs joins, foreign keys, unique constraints, transactions, JSONB, row-level triggers and a least-privilege database role. The triggers enforce append-only facts; the JSONB handles variable blocks and audit payloads. SQLite would be simpler for a demo, but it gives a weaker concurrency, role and TLS story. A document database would make the review queries and constraints harder.

Synthetic data and safety rules

Use only invented data. These rules are part of the assignment.

• **Plausible, not perfect.** Labour records should be clinically coherent — times in order, observations at sensible intervals, alerts followed by an assessment — and should include realistic imperfections such as a missed observation. They do not need medical review. If an AI tool generates them, check them yourself.
• **Variety.** Cover the cases listed under Essential requirement 13. A brought-dead case is a good test if you have time.